

Data and Artificial Intelligence

Cyber Shujaa Program

Week 9 Assignment

Bonus: Streamlit app Integration

Student Name: Salome Kungu

Student ID: CS-DA03-26054

Contents

Data and Artificial Intelligence	1
Cyber Shujaa Program.....	1
Week 9 Assignment Bonus: Streamlit app Integration	1
Introduction	2
The key objectives of the assignment were to:.....	2
Tasks Completed	2
Task Streamlit Model Deployment Dashboard	2
Screenshot of Streamlit Application	4
Screenshots of VS Code:	5
Conclusion.....	7

Introduction

The objective of this bonus task was to integrate the trained machine learning model into an interactive **Streamlit web application**. The purpose of the deployment was to demonstrate how a trained model can be transformed from a development environment into a simple and accessible application that allows users to interact with the model and generate predictions.

The Streamlit application loads the previously saved machine learning pipeline (**california_knn_pipeline.pkl**) and provides an interface where users can input housing features and receive predicted housing prices in real time.

This step represents an important stage in the machine learning workflow because it bridges the gap between **model development and practical usage**. Instead of running predictions through code in a notebook, the model becomes accessible through a user-friendly interface.

The deployment also demonstrates a fundamental concept in **Machine Learning Operations (MLOps)**, where trained models are integrated into applications that can be used by different users without requiring knowledge of programming.

The key objectives of the assignment were to:

1. Load a saved ML model (.pkl file) into a Streamlit app.
2. Create an intuitive UI for users to input data.
3. Display predictions and model performance metrics interactively.
4. (Optional) Add visualizations like feature importance or data insights.

Tasks Completed

Link to Code via Streamlit:

<https://kijosal-dev-mlops-streamlit-app-integration-app-dwqgqs.streamlit.app/>

Link to Code via GitHub:

<https://github.com/KijoSal-dev/MLOPs-Streamlit-app-Integration>

The following tasks were completed during the Streamlit deployment process:

1. Loaded the saved trained machine learning pipeline using pickle.
2. Built an interactive Streamlit web interface to collect housing feature inputs from users.
3. Connected the input data to the trained model pipeline for prediction.
4. Displayed predicted housing prices dynamically within the web application.
5. Implemented a simple visualization to provide insights into the dataset.
6. Tested the application locally to ensure accurate predictions and functionality.

These tasks demonstrate how machine learning models can be deployed in a simplified dashboard that allows real-time predictions.

Task Streamlit Model Deployment Dashboard

What:

In this task, the trained machine learning pipeline (**california_knn_pipeline.pkl**) was integrated into a **Streamlit application** to create a simple interactive prediction dashboard.

The application loads the trained K-Nearest Neighbors regression model and allows users to input the following housing features:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

After entering the required information, the user can click a **prediction button** to generate an estimated housing price based on the trained model.

The dashboard also includes a simple visualization that provides insight into the distribution of selected dataset features.

Why:

The purpose of deploying the model using Streamlit was to demonstrate how machine learning models can be transformed into **interactive applications that allow users to make predictions without directly interacting with the code**.

Streamlit simplifies the deployment process by enabling developers to build web applications using only Python code. This makes it possible to quickly prototype and share machine learning solutions.

This deployment also highlights the transition from **model experimentation to practical implementation**, which is a key concept in MLOps workflows.

Through this task, the trained model becomes accessible through a graphical interface where users can explore different input scenarios and observe the predicted outcomes.

Key Features of the Streamlit Dashboard

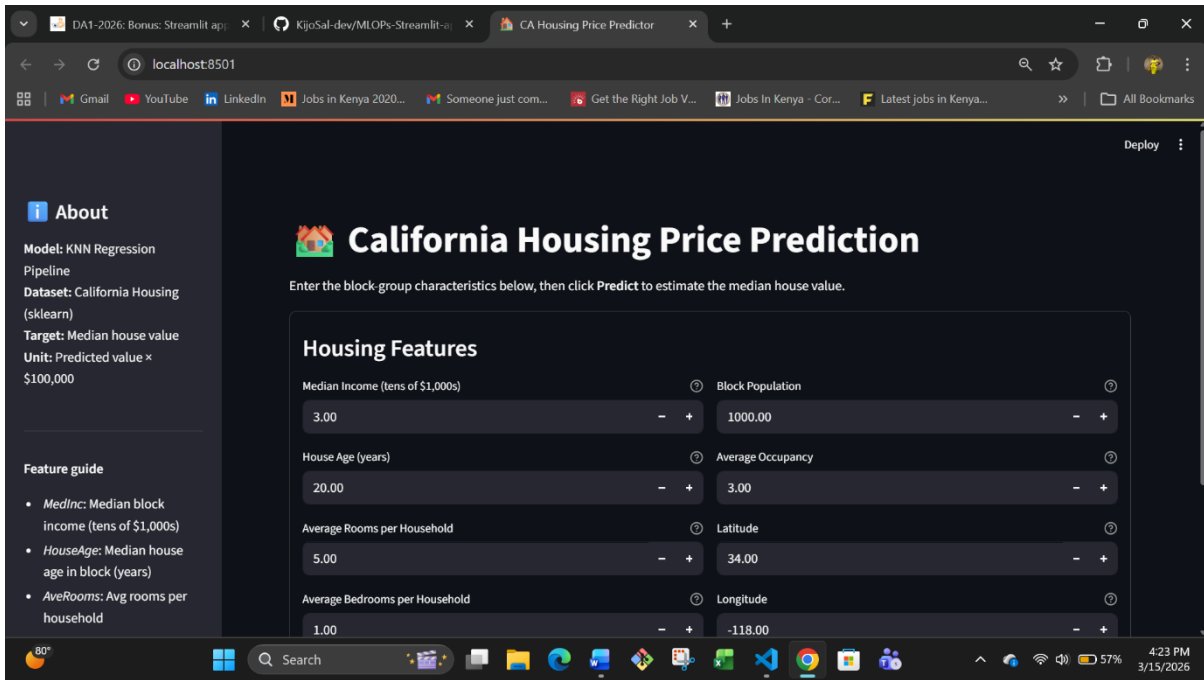
The application includes several key features:

1. **Interactive Input Fields**
Users can input housing feature values using number input fields.
2. **Prediction Button**
A button triggers the trained model to compute predictions.
3. **Predicted Housing Price Display**
The predicted housing price is clearly displayed in the interface.

4. Data Visualization

A simple chart displays feature distribution to provide additional insights into the dataset.

Screenshot of Streamlit Application



About

Model: KNN Regression Pipeline
 Dataset: California Housing (sklearn)
 Target: Median house value
 Unit: Predicted value × \$100,000

Feature guide

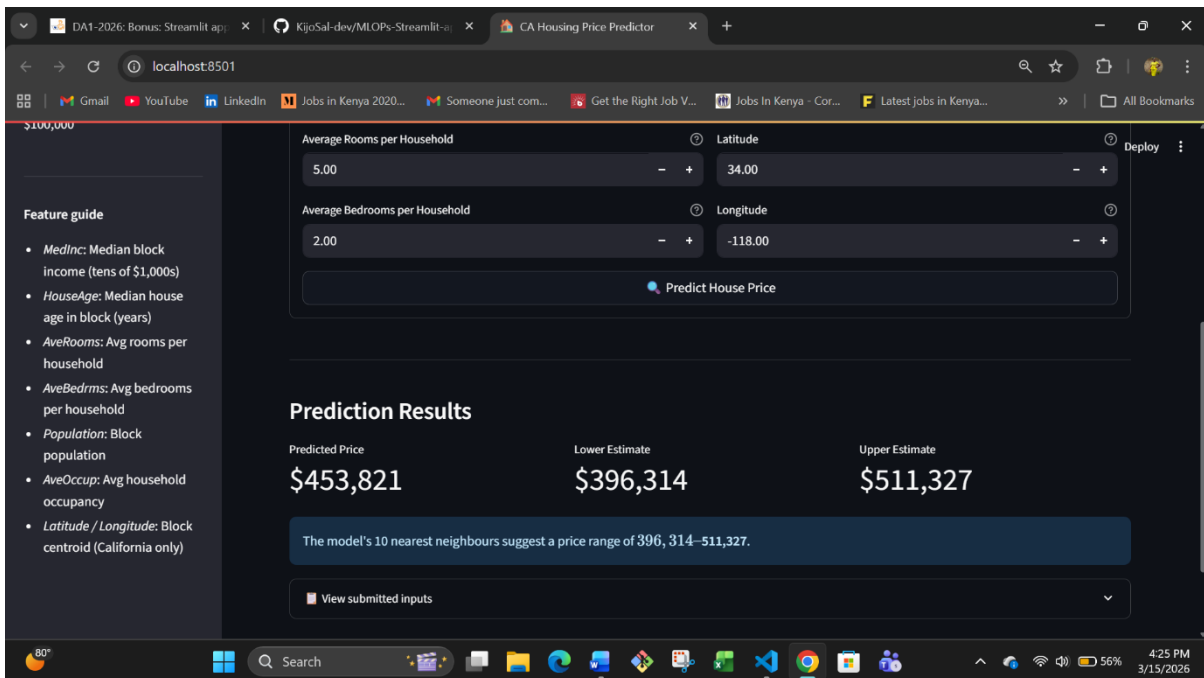
- *MedInc*: Median block income (tens of \$1,000s)
- *HouseAge*: Median house age in block (years)
- *AveRooms*: Avg rooms per household

California Housing Price Prediction

Enter the block group characteristics below, then click Predict to estimate the median house value.

Housing Features

Median Income (tens of \$1,000s)	3.00	Block Population	1000.00
House Age (years)	20.00	Average Occupancy	3.00
Average Rooms per Household	5.00	Latitude	34.00
Average Bedrooms per Household	1.00	Longitude	-118.00



Feature guide

- *MedInc*: Median block income (tens of \$1,000s)
- *HouseAge*: Median house age in block (years)
- *AveRooms*: Avg rooms per household
- *AveBedrms*: Avg bedrooms per household
- *Population*: Block population
- *AveOccup*: Avg household occupancy
- *Latitude / Longitude*: Block centroid (California only)

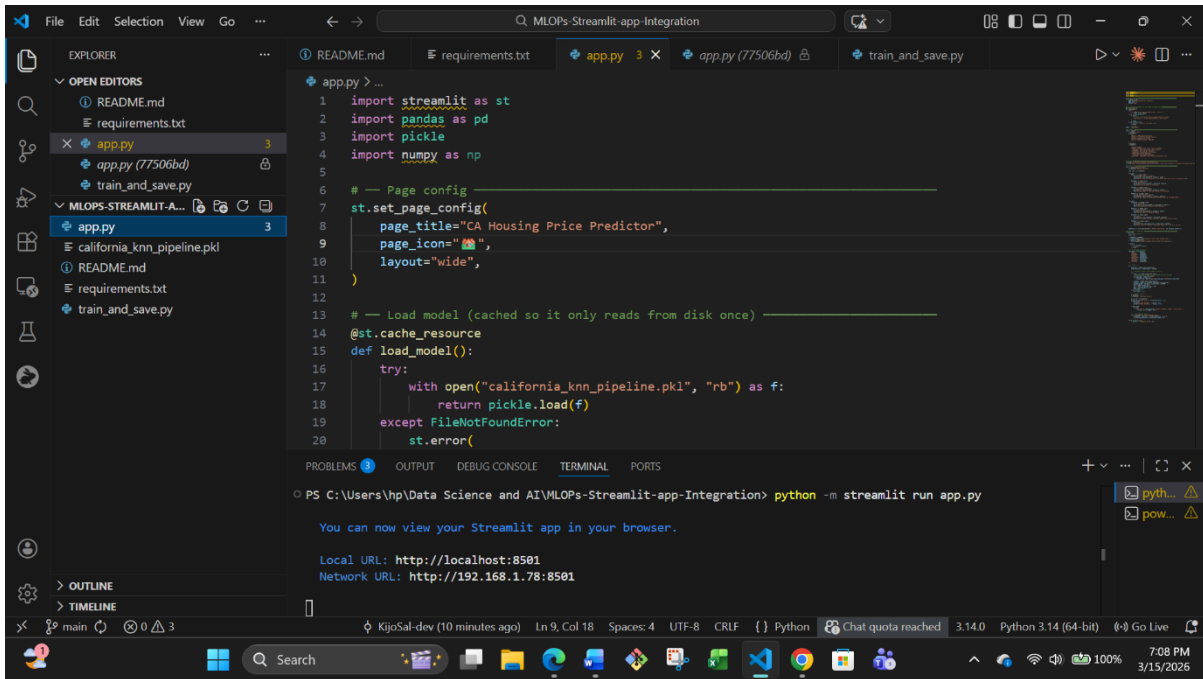
Prediction Results

Predicted Price	Lower Estimate	Upper Estimate
\$453,821	\$396,314	\$511,327

The model's 10 nearest neighbours suggest a price range of 396,314–511,327.

[View submitted inputs](#)

Screenshots of VS Code:



```

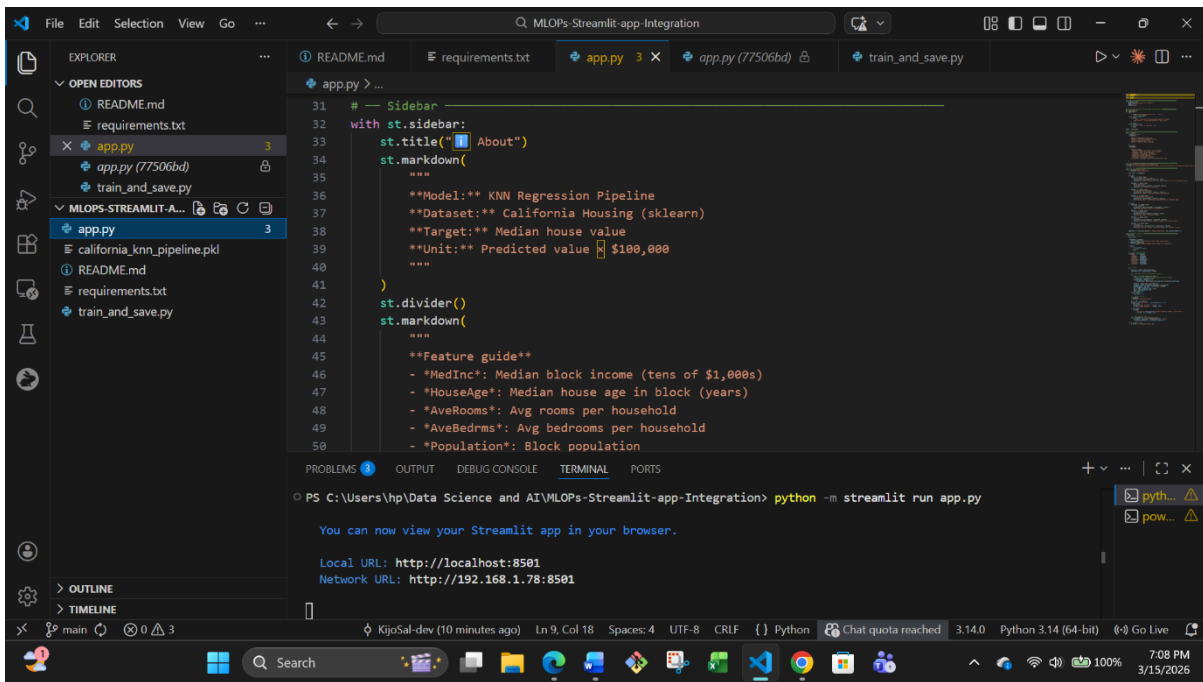
1 import streamlit as st
2 import pandas as pd
3 import pickle
4 import numpy as np
5
6 # -- Page config
7 st.set_page_config(
8     page_title="CA Housing Price Predictor",
9     page_icon="🏠",
10    layout="wide",
11)
12
13 # -- load model (cached so it only reads from disk once)
14 @st.cache_resource
15 def load_model():
16     try:
17         with open("california_knn_pipeline.pkl", "rb") as f:
18             return pickle.load(f)
19     except FileNotFoundError:
20         st.error(

```

PS C:\Users\hp\Data Science and AI\MLOPs-Streamlit-app-Integration> python -m streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>
 Network URL: <http://192.168.1.78:8501>



```

31 # -- Sidebar
32 with st.sidebar:
33     st.title("🏠 About")
34     st.markdown(
35         """
36         **Model:** KNN Regression Pipeline
37         **Dataset:** California Housing (sklearn)
38         **Target:** Median house value
39         **Unit:** Predicted value $100,000
40         """
41     )
42     st.divider()
43     st.markdown(
44         """
45         **Feature guide**
46         - *MedInc*: Median block income (tens of $1,000s)
47         - *HouseAge*: Median house age in block (years)
48         - *AveRooms*: Avg rooms per household
49         - *AveBedrms*: Avg bedrooms per household
50         - *Population*: Block population

```

PS C:\Users\hp\Data Science and AI\MLOPs-Streamlit-app-Integration> python -m streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>
 Network URL: <http://192.168.1.78:8501>

File Edit Selection View Go ...

Q MLOPs Streamlit-app-Integration

EXPLORER

- OPEN EDITORS
 - README.md
 - requirements.txt
 - app.py
 - app.py (77506bd)
 - train_and_save.py
- MLOPS-STREAMLIT-A...
 - app.py
 - california_knn_pipeline.pkl
 - README.md
 - requirements.txt
 - train_and_save.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

56 # -- Title
57 st.title("🏠 California Housing Price Prediction")
58 st.write("Enter the block-group characteristics below, then click **Predict** to estimate the median house
59
60 # -- Input form
61 with st.form("prediction_form"):
62     st.subheader("Housing Features")
63
64     col1, col2 = st.columns(2)
65
66     with col1:
67         MedInc = st.number_input(
68             "Median Income (tens of $1,000s)",
69             min_value=0.0, max_value=20.0, value=3.0, step=0.1,
70             help="Median income for households in the block, measured in tens of thousands of USD."
71         )
72         HouseAge = st.number_input(
73             "House Age (years)",
74             min_value=0.0, max_value=100.0, value=20.0, step=1.0,
75             help="House age in years."
76         )
77
78     with col2:
79         Population = st.number_input(
80             "Block Population",
81             min_value=1.0, max_value=40000.0, value=1000.0, step=10.0,
82             help="Total population living in the block group."
83         )
84         AveOccup = st.number_input(
85             "Average Occupancy",
86             min_value=1.0, max_value=20.0, value=3.0, step=0.1,
87             help="Average number of household members."
88         )
89         Latitude = st.number_input(
90             "Latitude",
91             min_value=32.5, max_value=42.0, value=34.0, step=0.01,
92             help="Block centroid latitude. California ranges from ~32.5°N to 42°N."
93         )
94         Longitude = st.number_input(
95             "Longitude",
96             min_value=-120.0, max_value=-114.0, value=-118.0, step=0.01,
97             help="Block centroid longitude. California ranges from ~-120°W to -114°W."
98         )
99
100     st.button("Predict")
101
102 if st.button("Predict"):
103     # Predict the median house value
104     # This is where the model prediction logic would go
105     # For now, we'll just return a placeholder value
106     median_house_value = 100000.0
107
108     st.write(f"Predicted Median House Value: {median_house_value}")
109
110 if __name__ == "__main__":
111     streamlit run app.py
  
```

PS C:\Users\hp\Data Science and AI\MLOPs-Streamlit-app-Integration> python -m streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>
Network URL: <http://192.168.1.78:8501>

main 0 3

KijoSal-dev (10 minutes ago) Ln 9, Col 18 Spaces: 4 UTF-8 CRLF {} Python Chat quota reached 3.14.0 Python 3.14 (64-bit) Go Live

7:08 PM 3/15/2026

File Edit Selection View Go ...

Q MLOPs Streamlit-app-Integration

EXPLORER

- OPEN EDITORS
 - README.md
 - requirements.txt
 - app.py
 - app.py (77506bd)
 - train_and_save.py
- MLOPS-STREAMLIT-A...
 - app.py
 - california_knn_pipeline.pkl
 - README.md
 - requirements.txt
 - train_and_save.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

87
88
89 with col2:
90     Population = st.number_input(
91         "Block Population",
92         min_value=1.0, max_value=40000.0, value=1000.0, step=10.0,
93         help="Total population living in the block group."
94     )
95     AveOccup = st.number_input(
96         "Average Occupancy",
97         min_value=1.0, max_value=20.0, value=3.0, step=0.1,
98         help="Average number of household members."
99     )
100     Latitude = st.number_input(
101         "Latitude",
102         min_value=32.5, max_value=42.0, value=34.0, step=0.01,
103         help="Block centroid latitude. California ranges from ~32.5°N to 42°N."
104     )
105     Longitude = st.number_input(
106         "Longitude",
107         min_value=-120.0, max_value=-114.0, value=-118.0, step=0.01,
108         help="Block centroid longitude. California ranges from ~-120°W to -114°W."
109     )
110
111     st.button("Predict")
112
113 if st.button("Predict"):
114     # Predict the median house value
115     # This is where the model prediction logic would go
116     # For now, we'll just return a placeholder value
117     median_house_value = 100000.0
118
119     st.write(f"Predicted Median House Value: {median_house_value}")
120
121 if __name__ == "__main__":
122     streamlit run app.py
  
```

PS C:\Users\hp\Data Science and AI\MLOPs-Streamlit-app-Integration> python -m streamlit run app.py

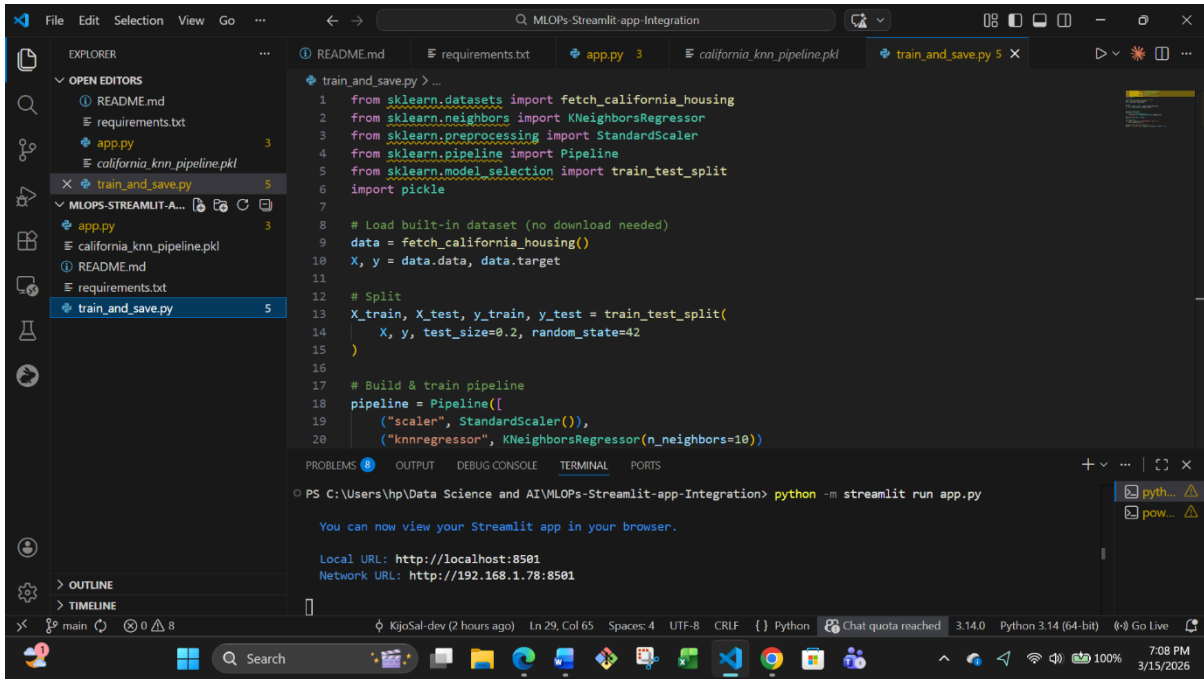
You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>
Network URL: <http://192.168.1.78:8501>

main 0 3

KijoSal-dev (10 minutes ago) Ln 9, Col 18 Spaces: 4 UTF-8 CRLF {} Python Chat quota reached 3.14.0 Python 3.14 (64-bit) Go Live

7:08 PM 3/15/2026



The screenshot shows a VS Code editor window titled "MLOPs Streamlit-app-Integration". The Explorer panel on the left shows the project structure with files like README.md, requirements.txt, app.py, california_knn_pipeline.pkl, and train_and_save.py. The main editor displays the content of train_and_save.py, which includes imports for sklearn datasets, neighbors, preprocessing, pipeline, and model selection, followed by code to load the dataset, split it, build a pipeline with a StandardScaler and KNeighborsRegressor, and save the pipeline. The TERMINAL panel at the bottom shows the command `python -m streamlit run app.py` being executed, with output indicating the app is running on `http://localhost:8501` and `http://192.168.1.78:8501`.

```

1 from sklearn.datasets import fetch_california_housing
2 from sklearn.neighbors import KNeighborsRegressor
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.pipeline import Pipeline
5 from sklearn.model_selection import train_test_split
6 import pickle
7
8 # Load built-in dataset (no download needed)
9 data = fetch_california_housing()
10 X, y = data.data, data.target
11
12 # Split
13 X_train, X_test, y_train, y_test = train_test_split(
14     X, y, test_size=0.2, random_state=42
15 )
16
17 # Build & train pipeline
18 pipeline = Pipeline([
19     ("scaler", StandardScaler()),
20     ("knnregressor", KNeighborsRegressor(n_neighbors=10))
21 ])
22
23 # Save the pipeline
24 with open('california_knn_pipeline.pkl', 'wb') as f:
25     pickle.dump(pipeline, f)
26
27 # Load the pipeline
28 with open('california_knn_pipeline.pkl', 'rb') as f:
29     loaded_pipeline = pickle.load(f)
30
31 # Predict using the loaded pipeline
32 loaded_pipeline.predict(X_test)

```

PS C:\Users\hp\Data Science and AI\MLOPs-Streamlit-app-Integration> python -m streamlit run app.py

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>
 Network URL: <http://192.168.1.78:8501>

Conclusion

This assignment provided practical experience in developing a complete machine learning workflow, from dataset exploration and preprocessing to model development, optimization, evaluation, and deployment.

The California Housing dataset was used to build a regression model capable of predicting median house values using several demographic and geographic features. The K-Nearest Neighbors regression model was implemented within a pipeline to ensure consistent preprocessing and prediction steps. Hyperparameter tuning using **GridSearchCV** improved the model's performance and ensured that the most suitable parameter configuration was selected.

Evaluation metrics such as **R² Score**, **Mean Squared Error**, and **Root Mean Squared Error** were used to assess the accuracy and reliability of the model.

Finally, the trained model pipeline was saved and integrated into deployment environments. A **FastAPI service** demonstrated how machine learning models can be exposed through APIs, while the **Streamlit application** provided an interactive interface that allows users to input housing features and receive predicted prices instantly.

Overall, this project demonstrated key concepts in machine learning development and **MLOps practices**, highlighting the process of moving models from experimentation to deployment in real-world applications.